RC Aircraft Bill of Materials

Electric single-motor high-wing trainer | span 1.1 m | hand launch or ground roll | cost-of-build only, nothing purchased

Motor type: BLDC (brushless DC). Rec. 1 is the A2212 1000KV outrunner and Rec. 2 the SunnySky X2212 980KV, both 3-phase brushless motors. They need a brushless ESC (3 motor wires, no brushes), which is why the ESC rows below are brushless-type. Brushless is chosen for efficiency, power-to-weight and no brush wear, versus a brushed motor that is heavier, less efficient and short-lived at this power.

Design assumptions

- High-wing trainer, span 1.1 m, chord 0.22 m, area ≈ 0.24 m² (24 dm²), AR ≈ 5. 6 mm foam board (Depron/XPS) fuselage and wing, carbon-rod spar.
- 3S LiPo (11.1 V nominal, 12.6 V full). Tractor motor, 4 channels (aileron, elevator, rudder, throttle), 2.4 GHz radio, hand launch with belly landing or small wheels.

Parts A and B: components, two options each, specs, quantity, cost, source

Component	Rec. 1 (budget)	Rec. 2 (performance)	Key specs	Qty	₹ R1	₹ R2	Source / link
Airframe	6 mm Depron/XPS foam board, free plan	EPO/EPP 1.1–1.2 m trainer kit	1.1 m span, ~260 g	1	1,500	3,000	Amazon.in “Depron 6 mm”, hobby shops
Reinforcement	6 mm carbon rod ×2 (1 m), 3 mm ply motor mount	8 mm carbon tube + CF strip	Spar bending stiffness	1 set	900	1,300	Robu.in, Amazon.in “carbon rod”
Motor (BLDC)	A2212 1000KV ✓	SunnySky X2212 980KV	3S, 12–25 A, 9–10″ prop	1	375	1,800	[1] Robu; R2 SunnySky India dealers
ESC (brushless)	30 A SimonK/BLHeli, 5 V BEC	Hobbywing Skywalker 40 A UBEC	Cont. current, BEC	1	1,000	2,800	[2] / [3]
Propeller	9×4.7 slow-fly (3-pack)	APC 9×6E or 10×4.7	Match to motor	1 + spare	300	700	Amazon.in, Robu.in “9047 prop”
Battery	Orange 3S 2200 mAh 30C ✓ (175 g)	Orange 3S 2200 mAh 40C ×2	11.1 V, ≥ 40 A cont.	1 / 2	1,900	4,200	[4] / [5]
Tx + Rx	FlySky FS-i6X + Rx (AFHDS 2A) ✓	RadioMaster Pocket (ELRS) + ELRS Rx	6+ ch, 2.4 GHz	1 set	4,800	11,000	[6] Robu; R2 RC shops
Servos	4 × SG90 (9 g)	4 × MG90S metal gear	~1.5–2 kg·cm	4	480	1,200	Robu.in, Robokits “SG90/MG90S”
Servo extensions	2 × 30 cm	Same + Y-lead	3-wire	2	200	300	Robu.in
Control hardware	1 mm wire pushrods, clevises, horns	Ball links, nylon horns		1 set	400	600	Amazon.in
Landing gear	3 mm wire gear, 45 mm wheels	Sprung gear	Optional (hand launch)	1	250	500	Robu.in
Connectors, wire	XT60 pair, 3.5 mm bullets, heat-shrink	Branded XT60, silicone wire	Rated ≥ 30 A	1 set	300	500	Robu.in
Adhesives	Hot glue, foam-safe CA, UHU Por	Same + 5-min epoxy		1 set	600	800	Hardware shop, Amazon.in
Misc	Velcro, zip ties, tape, screws, 4×AA (Tx)	Same		1 set	450	650	Local
TOTAL					**13,455**	**29,350**	

✓ = product page found on Robu.in (motor ₹375 was the listed price; the Orange 3S 2200 mAh was out of stock at the time, so check stock or use the 3300 mAh 25C). Other prices are estimates; add live links for rows showing only a store and search term. The Robu FlySky bundle ships with an FS-iA10B receiver (10 ch, pre-bound), which also works. Not in totals: LiPo charger, watt meter, LiPo bag (≈ ₹3,500 if you do not own them).

Part C: Justification

1. Motor–propeller match and thrust check

Quantity	Value	Basis
No-load rpm	1000 KV × 11.1 V ≈ 11,100 (12,600 at full charge)	rpm = KV × V
Loaded rpm (9×4.7)	≈ 9,000–10,000	~80–90% of no-load
Prop pitch speed	9,500 rpm × 4.7 in ≈ 19 m/s vs 12 m/s cruise	pitch speed must exceed flight speed
Current / power	≈ 12–15 A, ≈ 150 W	typical for 9×4.7 on 3S
Static thrust	≈ 700–850 g (Robu quotes ~700 g class for A2212 1000KV)	estimate, to be verified

- **How to check thrust:** mount the motor on a luggage/kitchen scale (or a pivot arm), prop facing away, plane restrained. Log thrust (g), current (A) and voltage with a watt meter at 50%, 75% and 100% throttle for under 30 s. Pass if current ≤ motor/ESC limit, thrust ≥ 0.8 × takeoff mass, and the motor is warm, not hot. Cross-check on eCalc.

2. ESC rating and electrical margin

- Peak ≈ 15 A (up to ~18 A with a 10×4.7 prop). 30 A ESC ⇒ 30/15 = **2.0× margin** (rule ≥ 1.5×); the 40 A option gives 2.7×.
- BEC: 4 servos ≈ 1.5–2 A peak plus receiver 0.1 A, within a 2 A (R1) or 3 A (R2) rating. Robu notes the Skywalker 40 A UBEC uses a linear BEC and runs a bit warm, so keep servo load light and ventilate the ESC.
- Voltage: 3S max 12.6 V is within ESC (2–3S/3–4S) and motor ratings. Brushless ESC type is required for the BLDC motor.

3. Battery selection and trade-offs

- 3S 2200 mAh 25–30C: continuous = 2.2 Ah × 30 = 66 A (25C: 55 A) against ≈ 15 A needed. Mass 175 g (Robu spec).
- 3000 mAh adds ≈ 60 g (wing loading ↑, climb ↓); 2S is too low in power for 9–10″ props; 4S needs a different motor/ESC and costs more.
- Endurance: 2.2 Ah × 80% usable = 1.76 Ah ÷ ≈ 6 A mixed-throttle average ≈ 17 min; plan 10–12 min with reserve. Store/charge LiPo safely in a bag.

4. Servo count and control surfaces

- **4 servos:** 2 ailerons (one per wing), 1 elevator, 1 rudder. Throttle goes through the ESC (no servo). Total 4 + throttle = 5 receiver outputs, within a 6-channel receiver. Optional: rudder-elevator-only flight saves 1 servo but reduces control.

5. Structural material

Material	Strength	Mass	Cost	Repairability	Verdict
Foam board (Depron/XPS)	Medium + CF spar	Light	Low	Excellent: glue/tape	**Chosen**
Balsa + ply	High, brittle	Lightest	Medium–high	Fair, skilled	Alternative
EPO/EPP kit	Good, crash-tolerant	Medium	High	Poor at home	Performance option

6. Compatibility checks

Check	Requirement	Our design	OK
Voltage	Pack ≤ ESC and motor rating	3S 12.6 V vs 2–4S ESC	✓
Current	ESC ≥ 1.5 × peak	30 A vs 15 A	✓
Connectors	Same pack-to-ESC type; motor leads match	XT60; 3.5 mm bullets	✓
BEC load	Servo peak ≤ BEC rating	≈ 2 A vs 2–3 A	✓
Motor mount / prop	Firewall fits motor; prop clears fuselage and ground	Check on build	✓
Servo fit and CG	Servos fit cutouts; CG 25–33% chord	Balance with battery position	✓
Tx–Rx protocol	Same system; bound; failsafe set	FlySky AFHDS 2A (i6X + iA6B/iA10B) or ELRS pair; bind, set throttle-off failsafe, range test	✓

Note: 2.4 GHz is licence-free in India, but follow local drone/RC flying rules. Always arm with throttle at zero and props clear.

Part D: Budget and Bonus

1–4. Total cost, configurations and choice

Config	Contents	Total ₹
Budget	All Rec. 1 parts, DIY foam airframe, FlySky Tx/Rx, 1 battery	≈ 13,455
Performance	All Rec. 2 parts, EPO-class/foam+film airframe, ELRS radio, 2 batteries, metal-gear servos	≈ 29,350
Chosen: Budget + 2 upgrades	MG90S servos (+720) and a spare prop set (+400)	**≈ 14,575**

Safe cost-saving opportunities

- Build the airframe from foam board and a free plan; reuse hot glue and tools you own.
- SG90 on the tail, metal-gear MG90S on ailerons where loads are higher. Buy props in packs.
- Buy the Tx/Rx as a bundle and share the charger with a teammate. **Never** cut corners on battery, ESC or connectors (fire and crash risk).

Choice for the competition and why

- **Budget + 2 upgrades.** First builds crash, and cheap parts are cheap to replace; the thrust check below passes on the budget motor.
- Performance config costs ≈ ₹14,775 more for marginal gain unless the competition scores speed or payload.

Bonus: takeoff mass and propulsion sufficiency

Item	g	Item	g
Foam airframe + glue	260	Receiver	15
Carbon spars	40	4 servos	36
Motor (A2212)	52	Pushrods, horns	20
ESC	25	Wiring, connectors	30
Prop	12	Landing gear	25
3S 2200 mAh battery (spec 175)	175	Tape, misc	20
Subtotal	**710**	**+10% margin = takeoff mass**	**≈ 781**

- **Wing loading:** 781 g ÷ 24 dm² ≈ **33 g/dm²** (trainer range 25–40).
- **Stall speed** ($C_{L,max}$ ≈ 1.2, ρ = 1.225): $V_s = \sqrt{(2W/(\rho S C_L))} = \sqrt{(2 \times 7.66 / (1.225 \times 0.24 \times 1.2))}$ ≈ **6.6 m/s** (24 km/h). Hand launch at ≈ 8–10 m/s is easy.
- **Thrust-to-weight:** static thrust 700–850 g vs 781 g gives T/W ≈ **0.9–1.1**. This is adequate for a light foam trainer but not generous, so confirm it on the scale (target ≥ 0.8).
- **Cruise 12 m/s:** q = 88 Pa, C_L ≈ 0.36, C_D ≈ 0.06 (foam airframe, fixed gear, with induced drag) ⇒ drag ≈ 1.3 N. Prop thrust at 12 m/s ≈ 3.5–4 N, so excess ≈ 2.5 N and climb angle ≈ asin(2.5/7.7) ≈ **15–19°**.
- **Conclusion:** propulsion is sufficient if measured static thrust ≥ ~700 g. If it is below that, change to the 980KV performance motor or trim weight (drop the landing gear, saving 25 g).

Sources

[1] Robu: A2212 1000 KV BLDC motor (₹375 listed)
https://aws.robu.in/product/a2212-1000-kv-bldc-brushless-dc-motor-for-drone-without-soldered-connector

[2] Robu: A2212 + SimonK 30 A ESC combo
https://aws.robu.in/?p=5933

[3] Robu: Hobbywing Skywalker-40A-UBEC (linear UBEC)
https://robu.in/?p=1504093

[4] Robu: Orange 3S 2200 mAh 30C (175 g)
https://robu.in/?p=23741

[5] Robu: Orange 3S 2200 mAh 40C
https://st.robu.in/?p=30983

[6] Robu: FlySky FS-i6X bundle (AFHDS 2A)
https://st.robu.in/?p=33428